

Comprehensive Guide to Franka Emika Panda Robotic Arm

Table of Contents

1. 1. Introduction to Collaborative Robotics
2. 2. Overview of Franka Emika Panda
3. 3. Technical Specifications
4. 4. System Components
5. 5. Workspace and Dimensions
6. 6. Installation & Environment
7. 7. Programming with Desk
8. 8. Safety & CE Compliance
9. 9. Applications in Industry and Research
10. 10. Conclusion & References

Appendix: Tool Changing Methods in Franka Emika Panda

Franka Emika Panda does not have built-in automatic tool changing capabilities, yet it supports seamless integration with external mechanical or magnetic tool changers. These tool changers enable the robot to automatically switch between tools during operations, improving workflow flexibility and automation.

☐ Commonly Compatible Brands:

- Robotiq
- SCHUNK (e.g., VERO-S NSR-A50-20)
- Zimmer Group (WWS Series)
- Zimmer Magnetic Tool Coupler (for non-electrical tools)

These tool changers can be mounted on the robot flange (ISO 9409-1-A50 standard), allowing the Panda arm to:

- Approach and pick up tools from a predefined tray or rack.
- Rotate or lock into the tool.
- Continue its assigned task autonomously.

☐ Approximate Tool Changing Time:

- Move to tool location: 1–2 seconds
- Engage/disengage tool changer: 1–2 seconds
- Return to working position: 1–2 seconds

✓Total Time: 3 to 6 seconds

Optimized setups can reduce this to 2–3 seconds using precise tool alignment stations.

☐ Technical Notes:

- For hands-free tool changes, use self-locking changers that can be actuated via robot motion.
- Tools requiring power or signals (IO) must use changers capable of transmitting electrical connections.
- The system must be configured in the software (e.g., ROS, MoveIt, or Desk Apps) to identify which tool is active and apply the correct parameters accordingly.

☐ Software Configuration Overview:

If using Franka Desk GUI:

- Define tools as custom grippers.
- Integrate tool switching in Desk Apps and create task workflows.

If using ROS or MoveIt:

- Define each tool in the URDF file.
- Handle tool switching through service calls or action servers.

- Monitor the active tool state through tool state topics.

🔒 Safety Tip:

Incorporate proximity sensors or limit switches to confirm successful tool engagement. This ensures safety during autonomous tool switching.

1. Introduction to Collaborative Robotics

Collaborative robotics, also known as cobotics, refers to the use of robots that are designed to work safely and efficiently alongside humans in shared workspaces. Unlike traditional industrial robots that require safety cages and operate in isolation, collaborative robots (cobots) are equipped with advanced sensors, force limitations, and intelligent control algorithms to ensure human safety during interaction.

🔑 Key Features of Collaborative Robots:

- **Safety**: Cobots monitor force and speed to avoid harming humans. Some have built-in torque sensors to detect contact.
- **Ease of Use**: They are designed to be user-friendly with intuitive programming interfaces.
- **Flexibility**: Cobots can be easily reprogrammed and repurposed for different tasks, making them ideal for dynamic environments.
- **Compact Design**: Their lightweight and space-efficient design allows them to be used in a wide variety of settings, including small workshops.

🌐 Applications Across Industries:

- **Manufacturing**: Assembly, packaging, quality inspection.
- **Healthcare**: Surgical assistance, rehabilitation devices.
- **Research and Education**: Safe platforms for experimentation and learning.
- **Agriculture**: Harvesting, plant monitoring, data collection.

Collaborative robots bridge the gap between full automation and human labor, enabling hybrid workflows that combine the precision and consistency of robots with the adaptability and judgment of humans. Their use is expanding rapidly in both industrial and non-industrial environments.

The Franka Emika Panda is one of the leading collaborative robotic arms known for its high sensitivity, ease of programming, and human-friendly design, making it a valuable asset in modern robotic applications.

2. Overview of Franka Emika Panda

The Franka Emika Panda is a lightweight, 7-DOF collaborative robotic arm designed for advanced manipulation tasks. With high-resolution torque sensors in all joints and a sensitive control

system, it mimics human-like dexterity, enabling safe and adaptive interactions in dynamic environments.

Key Attributes:

- 7 degrees of freedom (DoF)
- Payload capacity of up to 3 kg
- Repeatability of ± 0.1 mm (ISO 9283)
- Maximum reach of 855 mm
- Intuitive visual programming via the Desk platform

Its compact and elegant design makes it suitable for labs, research centers, and industrial automation. The robot is known for rapid deployment and safe coexistence with humans.

3. Technical Specifications

□ Arm:

- Degrees of freedom: 7
- Payload: 3 kg
- Max reach: 855 mm
- Repeatability: ± 0.1 mm
- Sensors: Torque sensors in all 7 joints
- Max joint speed: Up to $180^\circ/\text{s}$
- Max Cartesian velocity: 2 m/s

⚡ Controller:

- Voltage: 100–240 VAC
- Power: Max 600 W (avg. 300 W)
- Weight: ~7 kg
- Protection: IP20
- Interfaces: Ethernet, Arm connector, Power connector

✋ Hand (Gripper):

- Type: Parallel gripper
- Force: 70 N continuous, 140 N max
- Travel: 80 mm (50 mm/s per finger)
- Weight: 0.7 kg

□ Desk Platform:

- Browser-based interface
- Visual programming
- Workflow creation using apps

□ Environmental:

- Operating temp: +15°C to +25°C (typical)
- Extended temp: +5°C to +45°C
- Humidity: 20–80% (non-condensing)

4. System Components

The Panda system includes the following core modules:

1. **Robot Arm**: 7-DoF articulated lightweight manipulator.
2. **Control Unit**: 19" rack-mountable controller.
3. **Hand Module**: Optional parallel gripper for object manipulation.
4. **User Interface (Desk)**: Web-based software for programming and monitoring.
5. **Tool Flange**: Standard ISO 9409-1-A50 mounting for end-effectors.
6. **External I/O Support**: For integration with sensors and industrial systems.

Each component is designed for plug-and-play functionality to minimize setup time.

5. Workspace and Dimensions

The Panda's compact footprint and flexible arm allow efficient use in limited spaces.

- Workspace reach: 855 mm radius
- Side view and top view representations available in datasheet
- Mounting orientation: Upright
- Flange: ISO 9409-1-A50

This design enables the robot to cover a spherical workspace ideal for tasks like assembly, pick-and-place, and inspection.

6. Installation & Environment

Installation involves securing the robot upright using mounting bolts, connecting the arm to the controller, and powering the system via IEC 60320-C14 connector.

▣ Recommended Conditions:

- Temperature: 15–25°C typical (5–45°C extended)
- Humidity: 20–80% non-condensing
- Protection Rating: IP30 (arm), IP20 (controller)

Ensure stable surface and avoid environments with excessive dust or moisture.

7. Programming with Desk

The Franka Desk platform offers browser-based, intuitive, block-style programming.

Key Features:

- Drag-and-drop workflow design
- App-based task creation (e.g., Move, Grasp, Wait)
- Visual feedback and live control
- Supports beginner and advanced users

Additionally, Panda supports programming through ROS, MoveIt, and custom APIs for advanced robotic applications.

8. Safety & CE Compliance

Franka Panda follows EU safety guidelines and can be shipped as a CE-certified solution.

Safety Features:

- Torque sensors for force limitation
- Emergency stop integration
- External enabling/activation inputs

For CE out-of-the-box, a checklist is provided. Otherwise, users must perform a risk analysis and meet EN ISO 10218-1 and EN ISO 13849 standards.

9. Applications in Industry and Research

🔍 Research:

- Human-robot interaction studies
- Machine learning experimentation
- Dexterous manipulation training

🔍 Industry:

- Precision assembly
- Packaging and palletizing
- Electronics testing and inspection

🔍 Education:

- Safe for use in academic labs
- Easy for students to learn programming and control

Its adaptability makes Panda a popular choice for cross-domain robotic solutions.

10. Conclusion & References

The Franka Emika Panda robotic arm exemplifies the future of collaborative automation — a blend of flexibility, safety, and precision. Whether in a factory or a research lab, it empowers users to innovate with ease.

📄 For more information, visit: www.franka.de

📄 Additional Reading:

- Franka Emika Datasheet (2018)
- ROS & MoveIt Documentation
- ISO 9409-1 and ISO 10218-1 standards

Appendix B: Updates from Franka Production 3 Datasheet (v1.4, 2023)

This appendix includes updated technical and safety information from the Franka Production 3 datasheet (Release Version 1.4, May 2023). This model refines and builds upon previous versions like the Franka Emika Panda.

📄 Arm & Motion Updates:

- Degrees of Freedom: 7
- Payload: 3 kg
- Max Reach: 855 mm
- Protection Rating: IP40 (improved from IP30)
- Ambient Temperature: +5 °C to +45 °C
- Cartesian Speed: Up to 2 m/s
- Pose Repeatability: < ±0.1 mm (ISO 9283)
- Guiding Force: ~2.5 N
- Adjustable Translational Stiffness: 10–3000 N/m
- Adjustable Rotational Stiffness: 1–300 Nm/rad

📄 Joint Velocity Limits:

- A1–A4: 150 °/s
- A5–A7: 301 °/s

📄 Interfaces:

- Enhanced safety-rated inputs for external enabling and emergency stop
- Support for external fieldbuses: Modbus/TCP, OPC UA
- Integrated end effectors: 2-finger and vacuum gripper support

📄 Advanced Safety Functions (PL d / Cat. 3 Certified by TÜV SÜD):

- Emergency Stop (X3.1)
- External Enabling Device (X4)
- Hand-Guiding Mode
- Safety-rated Speed and Separation Monitoring
- Safe Cartesian/Joint Limits (SLP-C, SLS-C, SLP-J, SLS-J)
- SEEPO: Safe End Effector Power Off

▣ Performance Compliance:

- Safety functions meet EN ISO 10218-1:2011 and EN ISO 13849-1:2015
- PFH (Probability of Failure per Hour) $< 1 \times 10^{-7}$

▣ Workspace & Dimensions:

- Workspace Reach: 855 mm
- Full reach with flange to base: 1188 mm
- Footprint radius: R 805 to R 855 mm

This updated specification reinforces the system's reliability, enhanced safety integration, and greater control flexibility in real-time applications.

Appendix C: ESD Compliance and Testing Results (2023)

This appendix presents electrostatic discharge (ESD) compliance and testing results for the Franka Production 3 and Franka Research 3, based on measurements taken in April 2023.

▣ Key Test Parameters:

- Resistance to Ground (RGP) measured at contact surfaces and flange, according to DIN IEC/TR 61340-5-1 and DIN EN 61340-5-2.
- Electrostatic surface voltage measured after rubbing the robot surfaces, as well as during operation, in line with DIN EN 61340-5-1.

□ Test Conditions:

- Temperature: 24.2 °C
- Relative Humidity: 44%

▣ ESD Performance Summary:

- Resistance to ground: $< 1 \text{ G}\Omega$
- Electrostatic surface voltage: $< 125 \text{ V}$
- Electrostatic field: $< 5000 \text{ V/m}$
- Measurements were taken at 16 different points across the robot body.

✓ Compliance:

- The robot meets the requirements of DIN EN 61340-5-1 (2017 edition), provided that it does

not come into direct contact with ESD-sensitive electronics.

- The standard specifies a minimum distance of 25 mm between the robot and ESD-sensitive components.

□ Important Note on Fingertips:

- The default fingertips supplied with the Franka Hand must be replaced with conductive material suitable for ESD-sensitive environments.
- A fingertip configurator is available at franka.world/fingertip-configurator to help design compliant fingertips.

📌 Recommendation:

- Due to variability in environmental conditions, it is advised to perform a verifying ESD measurement after integration into ESD-sensitive setups.